Vigneshwar B

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Summary

Software Engineer with 4+ years of Python development experience, building AI-powered workflows, Retrieval-Augmented Generation (RAG) pipelines, and automation platforms in production. Hands-on with LangChain, LangSmith, vector databases (Chroma/Pinecone), embeddings, and Model Context Protocol (MCP) for building and integrating AI systems with enterprise workflows. Hands-on experience building APIs with FastAPI and exposure to system design considerations and trade- offs for scalable services. Passionate about applying Agentic AI and LLMs (GPT, Claude) to solve real enterprise problems.

Skills

Languages	Python (primary/strong), C/C++, Java (basic)
AI / Agentic AI	LangChain, LangGraph, LangSmith, RAG, MCP, Vector Databases (Chroma, Pinecone), Embeddings, Prompt Engineering, LLM Integration (GPT/Claude/open-source)
Backend & APIs	Django, Flask, FastAPI, REST API development
Infra / DevOps	Docker, Kubernetes, Git, Linux, CI/CD, Splunk
Data & Databases	pandas, NumPy, MongoDB, MySQL
Other	OAuth2, Streamlit, Kafka (basic)

Work Experience

Cisco, Bangalore **Aug 2022 - Present**
Software Engineer

- Built a hybrid Retrieval-Augmented Generation (RAG) system over enterprise data using Python, LangChain, and a Chroma vector database — combining semantic vector search with metadata filtering, Reciprocal Rank Fusion (RRF) reranking, an LLM-as-router agent for multi-intent query decomposition, LLM-as-judge evaluation of answer quality, and anti- hallucination guardrails.
- Built and self-hosted an MCP (Model Context Protocol) server (FastMCP, Python, OAuth-enabled) exposing tools for Webex messaging, meeting scheduling, and check-in summarization — enabling LLM/agent clients to drive workflows via the standardized MCP interface.
- Developed and containerized a Django-based network device management platform with FastAPI-based internal services, SSH tunneling, persistent session pooling, and real-time monitoring — reducing manual execution effort by 80%.
- Reviewed code developed by other engineers, providing feedback on style, correctness, testability, and efficiency; led design reviews with peers and stakeholders to choose technologies and architectures for engineering frameworks; triaged and debugged production system issues, analyzing root cause and impact on service operations; authored and maintained technical documentation for internal engineering platforms.
- Designed and built an AI-powered workflow that automates an end-to-end test process — script generation, execution orchestration, log analysis, and automated report generation — cutting manual effort by 60%+ across the organization.
- Actively uses GitHub Copilot and prompt engineering for code generation, refactoring, and design exploration across AI-driven workflows.

Projects

- **Employee RAG — Hybrid RAG System (2026):** Architecture: hybrid retriever (semantic + metadata-filtered) fused via RRF reranking, query decomposition for multi-intent questions, in-session conversational memory, and an LLM-as-judge evaluation loop for answer quality — deployed with a Streamlit UI on top of a local Claude-compatible LLM gate for privacy. Tech: Python, LangChain, Chroma, RAG, LLMs, embeddings, Streamlit.
- **Webex MCP Server (2025):** Architecture: FastMCP-based MCP server with OAuth-secured tool schemas for Webex messaging, meeting scheduling, and check-in summarization, self-hosted on Cisco's internal IDE platform for direct LLM/agent client integration. Tech: Python, FastMCP, MCP, OAuth, Webex APIs, LLMs.

Education

Sri Sairam Engineering College **Aug 2018 - Jun 2022**
B.Tech. in Information Technology — CGPA: 8.88/10

Awards and Certificates

- **Rising Star and Automation Genie award:** Recognized across multiple quarters for developing reusable automation scripts with best coding practices and consistently delivering high-quality solutions on time for customers.
- **AI-Tech Practitioner**, **Python** Specialization & Machine Learning/Deep Learning Specialization (Coursera).
- **Content creator, YouTube (InfiTamil):** Published technical tutorials and coding challenges for the learning community.